

Experiment No 02

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Subject : Artificial Intelligence

Class : Final year

Q. Write a program to implement Eight Queens problem by using backtracking.

Code :

```
import tkinter as tk
```

```
N = 8
```

```
def is_safe(board, row, col):
```

```
    for i in range(row):
```

```
        if board[i][col] == 1:
```

```
            return False
```

```
    for i, j in zip(range(row, -1, -1), range(col, -1, -1)):
```

```
        if board[i][j] == 1:
```

```
            return False
```

```
    for i, j in zip(range(row, -1, -1), range(col, N)):
```

```
        if board[i][j] == 1:
```

```
            return False
```

```
    return True
```

```
def solve_n_queens(board, row):
```

```
    if row >= N:
```

```
        return True
```

```
    for col in range(N):
```

```
        if is_safe(board, row, col):
```

```
            board[row][col] = 1
```

```

        if solve_n_queens(board, row + 1):
            return True

        board[row][col] = 0

    return False

def initialize_board():
    return [[0 for _ in range(N)] for _ in range(N)]

def display_solution(board):
    window = tk.Tk()
    window.title("8 Queens Solution")
    window.geometry("400x400")

    for row in range(N):
        for col in range(N):
            color = "white" if (row + col) % 2 == 0 else "gray"
            cell = tk.Frame(window, width=50, height=50, bg=color)
            cell.grid(row=row, column=col)

            if board[row][col] == 1:
                queen = tk.Label(cell, text="♔", font=("Arial", 24), fg="black")
                queen.place(relx=0.5, rely=0.5, anchor="center")

    window.mainloop()

if __name__ == "__main__":
    board = initialize_board()
    if solve_n_queens(board, 0):
        display_solution(board)
    else:
        print("No solution exists")

```

Output:



Questions :-

1. What is the primary objective of solving the 8 Queens Problem in this context?
2. What are the key constraints in the 8 Queens Problem?
3. What role does backtracking play in solving the 8 Queens Problem?
4. How are diagonals checked for conflicts in the 8 Queens Problem?
5. What is the significance of constraint satisfaction problems like the 8 Queens Problem in computer science?